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CLINICAL RESEARCH ARTICLE



The relationship between childhood trauma, rs1360780 genotypes, *FKBP5* intron 7 methylation and posttraumatic stress disorder in women who have experienced rape

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ABSTRACT

Background: Posttraumatic stress disorder (PTSD) is a common sequela of rape. Dysfunction of the hypothalamic–pituitary–adrenal (HPA) axis, a core regulator of the stress response, has been implicated in the aetiology and chronicity of PTSD. FK506 binding protein (*FKBP5*) is a co-chaperone and functional regulator of the glucocorticoid receptor and the HPA-axis.

Objective: This study investigated main and interaction effects of childhood trauma and the *FKBP5* rs1360780 genotype on longitudinal *FKBP5* intron 7 methylation, and whether change in *FKBP5* methylation over time was associated with PTSD symptom severity over time.

Method: Women who experienced rape ($n = 96$) were recruited from post-rape care services in KwaZulu Natal, South Africa. Total PTSD symptom scores, derived from the Davidson Trauma Scale, were assessed at baseline, 3-months and 6-months post-rape. Methylation levels at five *FKBP5* intron 7 CpG sites were determined using EpiTYPER Sequenom MassArray technology. Genotyping of rs1360980 was completed using the Agena MassArray genotyping system. Mixed linear regression models were used to analyse the data.

Results: The interaction between rs1360780 genotype and childhood trauma was a significant predictor of *FKBP5* methylation over time. There was a significant positive correlation between childhood trauma and methylation levels in participants with the CT and TT genotypes, while there was a significant negative correlation between childhood trauma and methylation in CC genotype carriers. *FKBP5* methylation was not a predictor of PTSD scores over time.

Conclusion: This is the first study to investigate longitudinal change in *FKBP5* methylation in a demographically homogenous same-trauma sample. The findings implicate childhood trauma and *FKBP5* rs1360980 genotype in the trajectory of *FKBP5* methylation levels in the aftermath of rape. Further research is needed to investigate the longitudinal role of *FKBP5* intron 7 methylation in relation to PTSD symptom trajectories post-rape.

La relación entre el trauma infantil, los genotipos rs1360780, la metilación del intrón 7 *FKBP5* y el trastorno de estrés postraumático en mujeres que han experimentado una violación

Antecedentes: El trastorno de estrés postraumático (TEPT) es una secuela común de la violación. La disfunción del eje hipotalámico-hipofisario-adrenal (HHA), un regulador clave en la respuesta al estrés, ha sido implicada en la etiología y cronicidad del TEPT. La proteína de unión FK506 (*FKBP5*) es una co-chaperona y reguladora funcional del receptor de glucocorticoides y del eje HHA.

Objetivo: Este estudio investigó los efectos principales e interacciones entre el trauma infantil y el genotipo rs1360780 de *FKBP5* sobre la metilación longitudinal del intrón 7 de *FKBP5*, y si el cambio en la metilación de *FKBP5* a lo largo del tiempo se asociaba con la severidad de los síntomas de TEPT a lo largo del tiempo.

Método: Mujeres que experimentaron una violación ($n = 96$) fueron reclutadas de servicios de atención post-violación en KwaZulu Natal, Sudáfrica. Los puntajes totales de síntomas de TEPT, derivadas de la Escala de Trauma de Davidson, se evaluaron al inicio, a los 3 meses y a los 6 meses posteriores a la violación. Los niveles de metilación en cinco sitios CpG del intrón 7 de *FKBP5* se determinaron utilizando la tecnología EpiTYPER Sequenom MassArray. La genotipificación de rs1360780 se realizó mediante el sistema de genotipificación Agena MassArray. Se utilizaron modelos de regresión lineal mixta para analizar los datos.

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KEYWORDS

FK506 binding protein (*FKBP5*); posttraumatic stress disorder (PTSD); childhood trauma; Rs1360980 genotype; methylation; epigenetics

PALABRAS CLAVE

Proteína de unión FK506 (*FKBP5*); trastorno de estrés postraumático (TEPT); trauma infantil; genotipo rs1360780; metilación; epigenética

HIGHLIGHTS

- In this sample of women who experienced rape, the interaction between *FKBP5* rs1360780 genotypes and childhood trauma was a significant predictor of *FKBP5* intron 7 methylation over time.
- There was a significant positive correlation between childhood trauma and *FKBP5* methylation levels in participants with the rs1360780 CT and TT genotypes.
- *FKBP5* methylation was not a significant predictor of PTSD scores over time.

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Resultados: La interacción entre el genotipo rs1360780 y el trauma infantil fue un predictor significativo de la metilación de *FKBP5* a lo largo del tiempo. Hubo una correlación positiva significativa entre el trauma infantil y los niveles de metilación en participantes con los genotipos CT y TT, mientras que se observó una correlación negativa significativa entre el trauma infantil y la metilación en portadoras del genotipo CC. La metilación de *FKBP5* no fue un predictor de las puntuaciones de TEPT a lo largo del tiempo.

Conclusión: Este es el primer estudio que investiga el cambio longitudinal en la metilación de *FKBP5* en una muestra demográficamente homogénea expuesta al mismo tipo de trauma. Los hallazgos implican al trauma infantil y al genotipo rs1360780 de *FKBP5* en la trayectoria de los niveles de metilación de *FKBP5* tras una violación. Se necesita más investigación para explorar el papel longitudinal de la metilación del intrón 7 de *FKBP5* en relación con la evolución de los síntomas de TEPT después de una violación.

1. Introduction

The prevalence of posttraumatic stress disorder (PTSD) in sexual assault survivors is higher than in survivors of other trauma types (Dworkin, 2020). Childhood trauma has been associated both with adult experiences of sexual assault and an increased risk for PTSD (Bryant-Davis et al., 2010; Campbell et al., 2008; Möller et al., 2014; Nishith et al., 2000; Ullman et al., 2007; Ullman et al., 2009). Sex differences have been widely documented, with women at an increased risk of developing PTSD compared to men (Christiansen & Berke, 2020) and women also more likely to be victims of sexual assault (Hetzel-Riggin & Roby, 2013).

The hypothalamic–pituitary–adrenal (HPA) axis is integral to the regulation of the stress response and returning the body to homeostasis after exposure to a traumatic stressor (Daskalakis et al., 2015). Dysfunction of the HPA-axis has been implicated in the aetiology and course of PTSD following trauma exposure (Klaassens et al., 2012). A growing body of literature suggests that genetic and epigenetic mechanisms, such as DNA methylation, may mediate the interplay between trauma and HPA-axis functioning, especially when the initial trauma occurs during childhood (Zannas et al., 2015).

Most genetic and epigenetic studies of PTSD have focused on genes that are directly or indirectly associated with regulation of the HPA-axis (Morrison et al., 2019). One of the most commonly investigated genes is the FK506 binding protein gene (*FKBP5*) (Morrison et al., 2019). *FKBP5* is a co-chaperone and important functional regulator of the glucocorticoid receptor (GR) (Binder, 2009). Intracellularly, cortisol binds to a GR complex in the cytoplasm and enters the cell nucleus, where it binds to glucocorticoid response elements (GREs) (Zannas et al., 2016). Binding of the GR complex to GREs upregulates *FKBP5* transcription (Binder, 2009; Zannas et al., 2016). *FKBP5*, in turn, binds to GRs in the cytoplasm and inhibits GR translocation to the nucleus, thereby creating an ultra-short intracellular negative feedback loop (Zannas et al., 2016). Prolonged stress and increased expression of *FKBP5* may result in long-term methylation changes in HPA-axis genes and a dysregulated stress response.

These changes may underlie the aetiology and trajectory of PTSD (Morrison et al., 2019).

Single-nucleotide polymorphisms (SNPs) may also contribute to dysfunction of the HPA-axis by adversely mediating dynamic methylation/demethylation of *FKBP5* as part of the regulation of the ultra-short negative feedback loop (Binder, 2009). Several *FKBP5* SNPs have been investigated in PTSD and childhood trauma; however, not all findings consistently implicate the same SNPs in PTSD (Binder et al., 2008; Mehta & Binder, 2012; Watkins et al., 2016; Xie et al., 2010). One SNP, rs1360780 (intron 2, 400 bp from a GRE), has been more consistently identified across studies (Binder et al., 2008; Kang et al., 2019; Klengel et al., 2013; Ramo-fernández et al., 2019; Watkins et al., 2016; Xie et al., 2010; Yehuda et al., 2015). The rs1360780 CT and TT genotypes have been linked to an increased risk for developing PTSD in African- and European-American samples (Klengel et al., 2013; Watkins et al., 2016), and increased *FKBP5* transcription, both in the absence and presence of GR binding to GREs (Klengel et al., 2013). Increased transcription of *FKBP5* promotes cortisol resistance, delays activation of the negative feedback loop of the HPA-axis and prolongs the fight-or-flight response (Matosin et al., 2018; Zannas et al., 2016).

The rs1360780 SNP has also been found to moderate the association between childhood trauma and PTSD since it is believed that prolonged increased circulating cortisol subsequent to childhood trauma may result in neuroendocrinological changes underlying the functioning of the HPA-axis, especially in the presence of at least one rs1360780 T allele (Wang et al., 2018). These changes may include methylation/demethylation of CpG sites in the *FKBP5* gene (Klaassens et al., 2012).

Methylation studies have mainly focused on an overlapping region in intron 7 of the *FKBP5* gene since it contains several glucocorticoid response elements (GREs) (Paakinaho et al., 2010). Carrying the T allele or CT/TT genotype of rs1360780 (Kang et al., 2019) and increased methylation in intron 7 was associated with a history of trauma exposure (Yehuda et al., 2016a) in studies investigating older participants (e.g. Korean Vietnam veterans and

holocaust survivors). Other studies investigating methylation and the interaction between childhood trauma and rs1360780 in younger populations found that increased childhood trauma was associated with decreased methylation in T allele carriers in the offspring of holocaust survivors (Yehuda et al., 2016a), in German women of childbearing age (Ramo-fernández et al., 2019) and in predominately Hispanic American women (Grasso, 2020). Increased childhood trauma also interacted with the rs1360780 T allele to predict increased PTSD symptom severity in a predominantly African American, mixed gender, civilian sample (Klengel et al., 2013). Here, a 12.3% decrease in methylation in T allele carriers was observed, while the CC genotype carriers showed increased methylation in the same region (Klengel et al., 2013). One longitudinal study found that children carrying the CT/TT genotype of rs1360780 and born to mothers reporting high post-traumatic stress symptoms, were more likely to report higher posttraumatic stress symptoms compared to CC genotype carriers (Pereira et al., 2021).

Only one study has investigated longitudinal change in *FKBP5* intron 7 methylation in relation to change in PTSD symptom scores. The study investigated treatment responders and non-responders (Caucasian American veteran sample), following a mindfulness-based stress reduction intervention. Non-responders showed decreased methylation compared to responders, who showed increased methylation in the same region (Bishop et al., 2018). There were no significant differences in methylation levels between responders and non-responders at baseline, and baseline PTSD symptom severity scores were not associated with methylation levels (Bishop et al., 2018).

No studies to date have investigated the relationship between childhood trauma, rs1360780, *FKBP5* intron 7 methylation and PTSD symptom scores over time in rape-exposed women of African ethnicity. In the present study, conducted in a South-African cohort of rape-exposed women in a low-income setting, we investigated if childhood trauma and rs1360780 had a main or interaction effect on longitudinal *FKBP5* intron 7 methylation, and whether change in *FKBP5* methylation over time was associated with change in PTSD symptom scores over time (while controlling for childhood trauma and rs1360780 genotype), using baseline (0-20 days post-rape), 3-months and 6-months post-rape data.

2. Methods

2.1. Design and setting

The study followed a longitudinal, prospective follow-up design. Samples and data were analysed from a

subset of rape-exposed women ($n = 96$) recruited from the Rape Impact Cohort Evaluation (RICE) study, a cohort study investigating the impact of rape on women's health and their use of health services in South Africa ($n = 854$). For a more detailed description of RICE's study methods, see Abrahams et al. (2017). Survivors of rape were recruited from four post-rape care services in and around the city of Durban, located in the KwaZulu Natal province of South Africa. These post-rape care services provided comprehensive emergency care, including access to police, counselling, and medical and forensic care. Rape-exposed women were informed of the study after presenting to and receiving treatment from the service. Interested participants were invited to the study site for enrolment.

2.2. Participants

Participants were enrolled into the study within 20 days of the index rape and a baseline interview was completed. Participants were between 18 and 40 years of age and of Black African ethnicity. In the parent study, participants were excluded if they were severely distressed or in urgent need of psychiatric treatment/hospitalisation, if they had an intellectual disability or were more than 14 weeks pregnant at enrolment. For this sub-study, additional exclusions were applied to control for factors that could influence methylation levels. Participants were excluded if they were pregnant or lactating (at any timepoint), if they had HIV seroconverted (at any timepoint), or if they met diagnostic criteria for PTSD at baseline due to a past traumatic event other than the rape. Also, only participants with complete clinical datasets and blood samples with sufficient DNA at all timepoints were considered for inclusion.

2.3. Procedure and ethical considerations

Participants were recruited and included in this sub-study consecutively, until the target number was reached. Upon enrolment researchers explained the study procedure, obtained informed consent, and completed the baseline clinical interview and assessments with participants. Assessments included a demographic questionnaire; self-report questionnaires measuring risk and protective factors for the development of PTSD; a rapid HIV antibody blood spot test; a human chorionic gonadotropin (hCG) urine pregnancy test; and blood collection for DNA analysis. Nursing staff undertook HIV testing, pregnancy testing and venous blood collection. The nursing staff also recorded prescription medication use and tobacco smoking status at each timepoint. A research assistant administered the demographic questionnaire, the clinical interview (under supervision of a registered

trauma counsellor or registered nurse), and assisted participants with the self-report measures. Participants were invited back to the study site to complete a 3-month and 6-month follow-up visit and all assessments completed at baseline were repeated. Ninety-six participants were selected for the sub-study based on the quality of DNA available from blood samples i.e. DNA concentration (>100 ng/ μ l) and purity (A_{260}/A_{280} ratio > 1.8 and A_{260}/A_{230} ratio > 2.0). Ethical approval to conduct the RICE study was obtained from the ethics committee at the South African Medical Research Council (SAMRC; EC019-10/2013) approval to conduct the sub-study was obtained from the Health Research Ethics Committee at Stellenbosch University (S16/08/146).

2.4. Clinical measures

2.4.1. The Mini International Neuropsychiatric Interview

The Mini International Neuropsychiatric Interview (MINI) is a structured psychiatric interview that screens for 16 DSM-IV psychiatric disorders including mood disorders, anxiety disorders, alcohol and drug dependence, psychosis, eating disorders and personality disorders (Sheehan et al., 1998). The MINI was used at baseline to screen for PTSD based on prior traumas other than the rape. The MINI has shown good reliability and validity in various settings (Lecrubier et al., 1997).

2.4.2. Davidson Trauma Scale (DTS)

The Davidson Trauma Scale (DTS) is a self-report questionnaire used to assess seventeen symptoms of PTSD in accordance with the DSM-IV PTSD criteria (Davidson et al., 1997). The DTS was administered at all timepoints and responses were measured on a 5-point Likert scale for symptom frequency ranging from 0 ('not at all') to 4 ('every day') and symptom severity ranging from 0 ('not at all distressing') to 4 ('extremely distressing'). An uploading error resulted in missing value sets for the symptom severity subscale of the DTS, which was corrected once it was identified. There were no missing values for the DTS frequency subscale. Missing DTS symptom severity values were imputed in the RICE study using a multiple imputation model while maintaining the multivariate normal distribution. The symptom frequency and severity scores were added together to produce a PTSD total score ranging between 0 and 136. A total score of 40 or more was considered indicative of clinically significant PTSD symptoms. The DTS has shown excellent discriminating power for distinguishing between participants with and without PTSD at a cut-point of 40 (Davidson et al., 1997). The DTS showed excellent reliability in this study at each timepoint with a

Cronbach alpha score of .92 at baseline, .91 at 3-months and .93 at 6-months post-rape.

2.4.3. Childhood Trauma Questionnaire Short Form (CTQ-SF)

A modified version of the Childhood Trauma Questionnaire- Short Form (CTQ-SF) was used to measure exposure to childhood trauma before the age of 18 years, at the baseline visit (Bernstein & Fink, 1998). The 14 items measuring childhood trauma centred around sexual abuse, physical abuse, emotional abuse, parental neglect and domestic violence. Responses were measured on a 4-point Likert scale ranging from 1 ('never') to 4 ('very often'). Higher scores indicated higher levels of exposure to childhood trauma. The modified CTQ-SF showed acceptable reliability in this study with a Cronbach alpha score of .75 at baseline.

2.4.4. Life Events Checklist (LEC)

A modified version of the LEC was used to measure lifetime exposure to different trauma types at the baseline visit (Weathers et al., 2013). The modified version of the LEC measured direct exposure to nine trauma types using a dichotomous 'yes/no' response. The trauma types measured were imprisonment, civil unrest/war, serious injury, being close to death, murder of a family member or friend, unnatural death of a family member or friend, murder of a stranger/s, robbed at gunpoint or knifepoint and kidnapped. The number of 'yes' responses were added together to yield a total score ranging from 0 to 9, indicating the cumulative lifetime trauma load of participants.

2.4.5. Alcohol Use Disorders Identification Test (AUDIT)

The Alcohol Use Disorders Identification Test-Concise (AUDIT-C) version was used to measure hazardous alcohol use at baseline (Saunders et al., 1993). The original AUDIT consists of 10-items with responses measured on a 5-point Likert scale and response options specific to each individual item. A score of three or more on the AUDIT-C (the first three items of the AUDIT) is considered indicative of hazardous drinking in women (Bush et al., 1998). The AUDIT-C has shown good reliability and validity in various settings (de Meneses-Gaya et al., 2009). The AUDIT-C showed good reliability in this study with a Cronbach alpha score of .83.

2.5. Experimental procedure

DNA was extracted from peripheral blood samples using the Gentra Puregene DNA extraction kit (Qiagen, Germany) and quantified by fluorimetry, using the Pico Green dsDNA quantitation reagent (ThermoFisher Scientific, MA, United States). Sample

concentrations were normalised to 50 ng/ul and shipped on ice to Inqaba Biotec in Pretoria, South Africa for genotyping, bisulfite conversion and methylation analysis.

Genotyping of rs1360780 was completed using target sequence detection on the Agena MassARRAY system (Agena Bioscience, CA, United States). Polymerase chain reaction (PCR) was followed by single base extension using dideoxynucleoside terminators of the SNP specific primer. Matrix-assisted laser desorption ionisation time-of-flight (MALDI-TOF) mass spectrometry was then used to identify SNP genotypes.

Methylation analysis was completed using the EpiTYPER Sequenom MassARRAY system (Agena Bioscience, CA, United States) technology. Here, PCR primer design was performed using Sequenom Epidesigner software (www.epidesigner.com). For quality control, six bisulphite-converted samples were tested in triplicate along with four non-converted samples and two water samples, the latter two were used as controls. The samples were randomly assigned to one of three plates i.e. 96 samples per plate, three plates to accommodate the samples collected at each timepoint. Samples were bisulphite-converted using the Zymo EZ DNA methylation kit (Zymo Research, California, United States). Bisulphite conversion was followed by PCR amplification of the target region using t7-promoter tagged primers (forward primer, 5'-aggaagagagTTGTTTTTGGAAATTTAAGGTAATTGA-3' and reverse primer, 5'-cagtaatacgactcactatagggaagaggctCCTCCAACACTACTACTAAAAAATAAAACA-3'). RNA transcription, base-specific RNA cleavage and finally DNA methylation quantification were completed using the SpectroCHIP array, MALDI-TOF mass spectrometry and EpiTYPER software. For the full laboratory procedure see Suchiman et al. (2015). Findings resulting from the EpiTYPER technology have proven to be highly reproducible and have been found to detect methylation levels as low as 5% (Claus et al., 2012; Ehrich et al., 2005).

We followed the same approach as Klengel et al. (2013) in selecting the *FKBP5* intron 7 CpG sites to investigate methylation differences (Klengel et al., 2013). Of the six CpG sites (divided into 3 bins to form putative functional units around transcription factor binding sites) investigated in Klengel et al. (2013), we investigated two CpG sites in bin 1 (Bin1-CpG1 and Bin1-CpG2) and two sites in bin 2 (Bin2-CpG1 and Bin2-CpG2-GRE). We could not investigate Bin2-CpG3 since the mass of silent peaks in the surrounding area precluded detection of methylation levels in this CpG site. We also could not investigate Bin3-CpG1-GRE since its mass was too low to be detected. We included an additional CpG site located 2 base pairs from the end of the

GRE in Bin3. We labelled the five CpG sites investigated in this study as CpG1-CpG5 (chr6: 35558310-35558785). Sequences of the CpG sites investigated are based on the Human Genome Build 37 (GRCh37/hg19).

2.6. Data analysis

Most variables did not conform to a normal distribution and non-parametric statistics were therefore used in univariate analysis. Descriptive statistics were calculated for baseline demographic and clinical variables. Baseline age, education, employment, relationship status, HIV status, smoking status, lifetime trauma exposure, and alcohol and medication use were investigated as potential confounding variables in relation to PTSD and methylation. Potential confounding/covarying variables were identified using Spearman's correlation coefficients, Chi-square tests and Mann-Whitney U test statistics. The relationship between PTSD scores and rs1360780 genotype was investigated using Mann-Whitney U test statistics. An online calculator and Chi-square test were used to determine if the observed rs1360780 genotype frequencies were consistent with Hardy-Weinberg equilibrium (<http://www.dr-petrek.eu/documents/HWE.xls>). Genotypes for rs1360780 were grouped into CC carriers and CT/TT carriers, given that the T allele and its adjacent alleles form a TATA box, which has been found to have a functional effect on gene expression (Matosin et al., 2018). The dichotomised rs1360780 variable (CC vs CT/TT) was used in all univariate and multivariate analyses.

Mixed linear regression models were used to investigate main and interaction effects of childhood trauma score and the rs1360780 genotype on CpG1-5 methylation change over time (baseline, 3-months and 6-months post-rape). Mixed regression models were also used to investigate the main effect of CpG1-5 methylation as a predictor of change in PTSD scores over time, with and without adjustment for childhood trauma score and rs1360780 genotype. An alpha value of 0.05 was used as an indicator of statistical significance. A Bonferroni-Holm correction ($p < .0042$) was applied to control for multiple testing in all analysis (univariate and multivariate), considering the time points ($n = 3$), CpG sites ($n = 5$), childhood trauma score ($n = 1$), genotype groups ($n = 2$), and PTSD scores ($n = 1$) investigated in the study.

3. Results

3.1. Demographic and clinical characteristics of the sample

Baseline demographic characteristics including (age, education, employment, relationship status, HIV

Table 1. Demographic and clinical characteristics of the sample.

	<i>n</i>	%	<i>M</i>	<i>SD</i>	Range
Age (baseline)	96	100	25.2	5.4	18–39
Basic education completed (baseline)	96	100			
No	39	40.6			
Yes	57	59.4			
Employed (baseline)	96	100			
No	74	77.1			
Yes	22	22.9			
Relationship status (baseline)	96	100			
Single	20	20.8			
In a relationship	76	79.2			
HIV status (baseline)	96	100			
Negative	50	52.1			
Positive	46	47.9			
Smoker (baseline)	96	100			
No	84	87.5			
Yes	12	12.5			
Medication use (baseline)	96	100			
No	64	66.7			
Yes	32	33.3			
Childhood trauma score (baseline)	96	100	16.7	3.4	14–29
Sexual abuse score	96	100	4.4	1.0	4–10
Physical abuse score	96	100	3.7	1.1	3–7
Emotional abuse score	96	100	2.5	1.1	2–8
Neglect score	96	100	4.8	1.3	4–11
Domestic violence score	96	100	1.3	0.6	1–4
Cumulative childhood trauma score	96	100	1.5	1.5	0–5
Lifetime trauma cumulative (baseline)	96	100	1.4	1.4	0–5
Imprisonment endorsed	3	3.1			
Civil unrest or war endorsed	4	4.2			
Serious injury endorsed	11	11.5			
Close to death endorsed	27	28.1			
Murder of family or friend endorsed	16	16.7			
Unnatural death of family or friend endorsed	14	14.6			
Murder of a stranger endorsed	15	15.6			
Robbed at gun- or knifepoint endorsed	35	36.5			
Kidnapped endorsed	7	7.3			
Cumulative lifetime trauma	96	100	1.4	1.4	0–5
Alcohol use score (baseline)	96	100	1.7	2.4	0–10
Not Hazardous	69	71.9			
Hazardous	27	28.1			
PTSD score (baseline)	96	100	66.4	20.1	10–104
CpG1 methylation (baseline)	84	100	93.0	10.9	31–100
CpG2 methylation (baseline)	84	100	81.9	8.0	55–100
CpG3 methylation (baseline)	84	100	90.2	10.6	37–99
CpG4 methylation (baseline)	85	100	93.7	6.8	63–100
CpG5 methylation (baseline)	85	100	65.6	8.9	44–87
PTSD score (3-months)	96	100	35.3	27.6	0–127
CpG1 methylation (3-months)	86	100	92.7	10.0	40–100
CpG2 methylation (3-months)	87	100	80.4	8.8	53–100
CpG3 methylation (3-months)	88	100	88.0	11.1	47–100
CpG4 methylation (3-months)	86	100	92.6	7.1	71–100
CpG5 methylation (3-months)	88	100	64.3	8.3	39–84
PTSD score (6-months)	96	100	26.0	28.0	0–133
CpG1 methylation (6-months)	84	100	92.2	11.3	42–100
CpG2 methylation (6-months)	83	100	80.9	7.7	61–99
CpG3 methylation (6-months)	84	100	89.5	8.2	56–100
CpG4 methylation (6-months)	83	100	92.9	6.4	66–100
CpG5 methylation (6-months)	84	100	62.7	9.4	34–86
rs1360980 genotype	95	100			
CT	31	32.3			
TT	52	54.7			
TT	12	12.6			

Abbreviations: Mean (*M*), standard deviation (*SD*), posttraumatic stress disorder (PTSD).

status, smoking status, medication use, childhood trauma exposure and lifetime trauma exposure) are presented in Table 1. PTSD scores and methylation percentages at baseline, and at 3- and 6-months are also included. The rs1360780 genotype frequencies were in Hardy-Weinberg equilibrium ($\chi^2 = 1.87$, $p = .171$) and are also reported in Table 1.

3.2. Childhood trauma and methylation

The correlation coefficients between childhood trauma scores and *FKBP5* methylation are presented in Table 2. There were no significant correlations between childhood trauma and methylation. The distribution of *FKBP5* methylation scores, considering a

Table 2. Correlation coefficients for the relationship between childhood trauma and methylation.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Childhood trauma (baseline)	1															
2. CpG1 methylation (baseline)	.025	1														
3. CpG2 methylation (baseline)	.047	.160	1													
4. CpG3 methylation (baseline)	-.042	.048	.404**	1												
5. CpG4 methylation (baseline)	-.141	.048	.473**	.259*	1											
6. CpG5 methylation (baseline)	.068	.276*	.272*	.163	-.021	1										
7. CpG1 methylation (3-months)	.075	.216	.127	.093	.242*	.098	1									
8. CpG2 methylation (3-months)	.031	.167	.291*	.128	.066	.317**	.016	1								
9. CpG3 methylation (3-months)	-.043	.151	.169	.006	-.093	.113	-.258*	.381**	1							
10. CpG4 methylation (3-months)	.010	.103	.266*	-.014	-.041	.065	-.179	.355**	.466**	1						
11. CpG5 methylation (3-months)	-.146	.272*	.198	.017	.137	.291**	.079	.156	.137	.246*	1					
12. CpG1 methylation (6-months)	-.129	.092	.112	-.135	.048	-.181	.160	-.007	.051	.028	.042	1				
13. CpG2 methylation (6-months)	-.128	.221	.072	.008	-.056	.300**	.075	.330**	-.039	.020	.164	-.163	1			
14. CpG3 methylation (6-months)	.140	.168	.169	.138	.082	.064	.069	.239*	.092	.137	.063	-.219*	.551**	1		
15. CpG4 methylation (3-months)	.005	.240*	-.013	-.064	-.063	.308**	.104	.063	0.025	-.004	.154	-.105	.416**	.364**	1	
16. CpG5 methylation (6-months)	-.164	.158	.204	.002	.106	.304**	.035	.059	.122	.061	.197	.127	.191	-.030	.182	1

median split of childhood trauma scores (Median = 16), is illustrated in Figure 1. There were no significant differences in methylation levels between those with low childhood trauma scores compared to those with high childhood trauma scores.

3.3. Rs1360780 genotype and methylation

There were no significant differences in *FKBP5* methylation scores between rs1360780 genotypes, except for CpG5 at baseline ($z = -1.98$, $p = .048$) where the CT/TT genotype showed lower methylation ($M = 64.12$, $SD = 8.86$) compared to the CC genotype ($M = 68.11$, $SD = 8.05$). The association did not survive the adjustment for multiple testing. The distribution of *FKBP5* methylation scores stratified by genotype, are illustrated in Figure 2.

3.4. PTSD and methylation

The correlations between PTSD scores and methylation scores are presented in Table 3. The distribution of *FKBP5* methylation scores, considering a median split of PTSD scores (baseline median = 66, 3-month median = 29, 6-month median = 16), is illustrated in Figure 3. There was a significant difference in baseline CpG2 methylation ($z = -2.13$, $p = .033$) between those with low PTSD scores ($M = 83.98$, $SD = 6.69$) and those with high PTSD scores ($M = 79.91$, $SD = 8.69$) at baseline. There were no significant differences between groups at 3-months. There was a significant difference in 6-month CpG4 methylation ($z = -2.33$, $p = .020$) between those with low PTSD scores ($M = 93.87$, $SD = 6.19$) and those with high PTSD scores ($M = 91.84$, $SD = 6.47$) at 6-months. The associations did not survive the adjustment for multiple testing.

3.5. Potential covariates

Associations between PTSD score, methylation levels, rs1360980 and potential covariates at baseline are presented in Tables 4 and 5. Baseline age, HIV status, smoking status, lifetime trauma exposure, alcohol use and medication use were not significantly associated with baseline PTSD scores, CpG1 – CpG5 methylation levels or rs1360780 genotype. There were no significant differences in PTSD scores between women who did and did not complete basic education. Unemployed women ($M = 69.35$, $SD = 18.78$) had significantly higher PTSD scores ($z = -2.68$, $p = .007$) compare to employed women ($M = 56.30$, $SD = 21.47$). Women who were not in a relationship ($M = 75.48$, $SD = 14.85$) had significantly higher PTSD scores ($z = -2.28$, $p = .022$) compare to women who were in a relationship ($M = 63.96$, $SD = 20.66$). The associations did not survive correction for multiple testing.

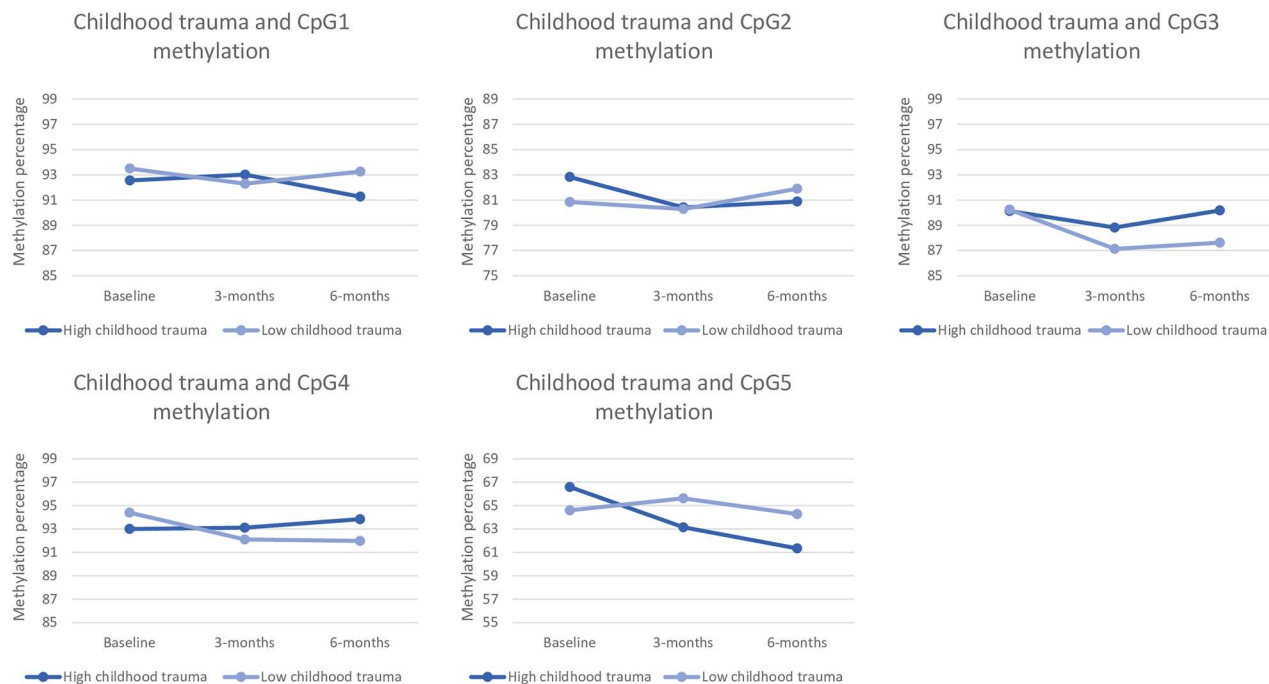


Figure 1. Childhood trauma and differences in methylation levels.

3.6. Main and interaction effects of childhood trauma and rs1360780 genotype on methylation trajectories post-rape

The main effect of childhood trauma as a predictor of *FKBP5* methylation levels over time was not significant for any of the CpG sites (see Table 6, models 1–5). The main effect of rs1360780 genotype as a predictor of *FKBP5* methylation levels over time was significant for CpG1 $\beta = .30.74$, $t(3.04)$, $p = .003$, CpG3 $\beta = -16.12$, $t(-2.42)$, $p = .017$ and CpG4 $\beta = -9.04$,

$t(-2.02)$, $p = .046$. Only the association between rs1360780 genotype and CpG1 methylation survived adjustment for multiple testing with lower methylation levels in participants with the CC genotype.

The interaction between rs1360780 genotype and childhood trauma was a significant predictor of CpG1 $\beta = -2.13$, $t(-3.62)$, $p = .0005$ and CpG3 $\beta = 0.86$, $t(2.19)$, $p = .030$ methylation over time. Only the association between rs1360780 genotype and childhood trauma interaction on CpG1 methylation

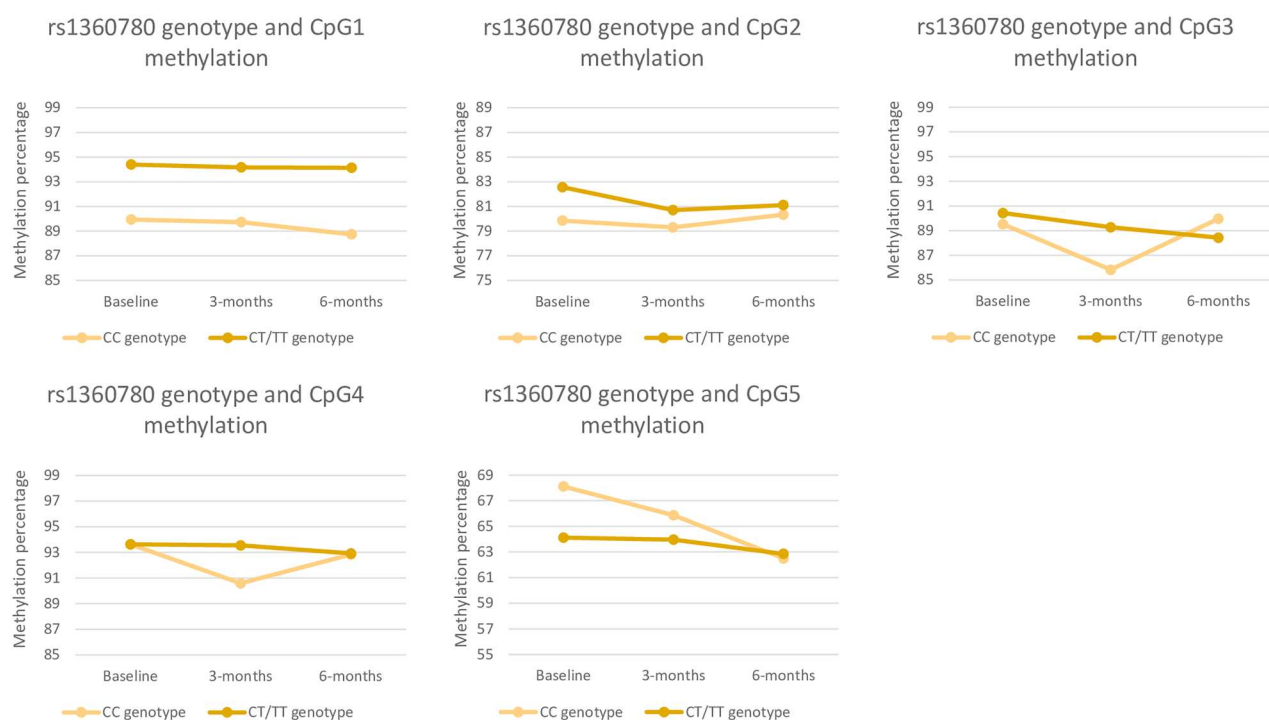


Figure 2. rs1360780 genotype and differences in methylation levels.

Table 3. Correlation coefficients for the relationship between PTSD and methylation.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. PTSD (baseline)	1																
2. PTSD (3-months)	.350**	1															
3. PTSD (6-months)	.143	.129	1														
4. CpG1 methylation (baseline)	-.095	-.145	-.007	1													
5. CpG2 methylation (baseline)	-.126	.092	.160	.160	1												
6. CpG3 methylation (baseline)	-.033	-.131	.166	.048	.404**	1											
7. CpG4 methylation (baseline)	-.074	.012	.053	.048	.473**	.259*	1										
8. CpG5 methylation (baseline)	.006	.048	-.150	.276*	.272*	.163	-.021	1									
9. CpG1 methylation (3-months)	-.061	-.116	.013	.216	.127	.093	.242*	-.021	.016								
10. CpG2 methylation (3-months)	-.027	-.020	.118	.167	.291*	.128	.066	.317**	-.258*	.381**	1						
11. CpG3 methylation (3-months)	-.104	.060	.024	.151	.169	.006	-.093	.113	-.179	.355**	.466**	1					
12. CpG4 methylation (3-months)	-.027	-.020	.118	.103	.266*	-.014	-.041	.065	.079	.156	.137	.246*	1				
13. CpG5 methylation (3-months)	-.104	.060	.823	.272*	.198	.017	.137	.291**	.160	-.007	.051	.028	.042	1			
14. CpG1 methylation (6-months)	.046	.016	.600	.092	.112	-.135	.048	-.181	.075	.330**	-.039	.020	.164	-.163	1		
15. CpG2 methylation (6-months)	.043	.068	-.118	.221	.072	.008	-.056	.300**	.069	.239*	.092	.137	.063	-.219*	.551**	1	
16. CpG3 methylation (6-months)	-.089	-.021	-.029	.168	.169	.138	.082	.064	.104	.063	.025	-.004	.154	-.105	.416**	.182	1
17. CpG4 methylation (6-months)	-.193	-.015	-.141	.158	.204	.002	.106	.304**	.104	.063	.025	-.004	.154	-.105	.416**	.182	1

survived adjustment for multiple testing. This interaction was driven by the CC genotype at 6-months post-rape. Lower childhood trauma scores were correlated with increased methylation levels in participants who carried the CC genotype at 6-months post-rape ($r = -.547$, $p = .002$), while the correlation between methylation and childhood trauma for those carrying the CT/TT genotype was not significant at the same timepoint (see Figure 4). The Fisher's Z test for comparison of correlation coefficients between the genotype groups was also significant, confirming that the effect was indeed driven by the CC genotype ($z = -3.22$, $p = .001$). The correlation coefficients between childhood trauma and CpG1 methylation over time, stratified by rs1360780 genotype, are presented in Figure 4.

3.7. Main and interaction effects of childhood trauma and rs1360780 genotype on PTSD symptom trajectory post-rape

Neither childhood trauma nor the rs1360780 genotype had a significant main effect on PTSD symptom trajectories over time. The interaction between childhood trauma and the rs1360780 genotype was also not a significant predictor of PTSD scores over time (see Table 4, model 6).

3.8. Change in methylation and PTSD scores over time

None of the *FKBP5* CpG sites were significant predictors of PTSD scores over time (see Table 6, model 7A-11A). The interaction between childhood trauma and the rs1360780 genotype was also no longer significant when *FKBP5* CpG site methylation was added to the models (see Table 6, model 7B-11B).

4. Discussion

The overarching aim of this study was to determine if childhood trauma score and rs1360780 genotype were independent predictors of longitudinal change in intron 7 *FKBP5* methylation levels as well as whether their interaction and *FKBP5* intron 7 methylation predicted change in PTSD scores over time. To our knowledge, this is the first study to longitudinally investigate the interaction between these variables in a rape-exposed African cohort.

After correction for multiple testing, childhood trauma, genotype and none of the CpG sites were independent significant predictors of PTSD in the regression models, although CpG1 methylation was nearing significance and may very well be a significant finding if the study were to be conducted in a larger sample (Suchiman et al., 2015). Higher methylation in CpG1 was associated with lower PTSD scores.

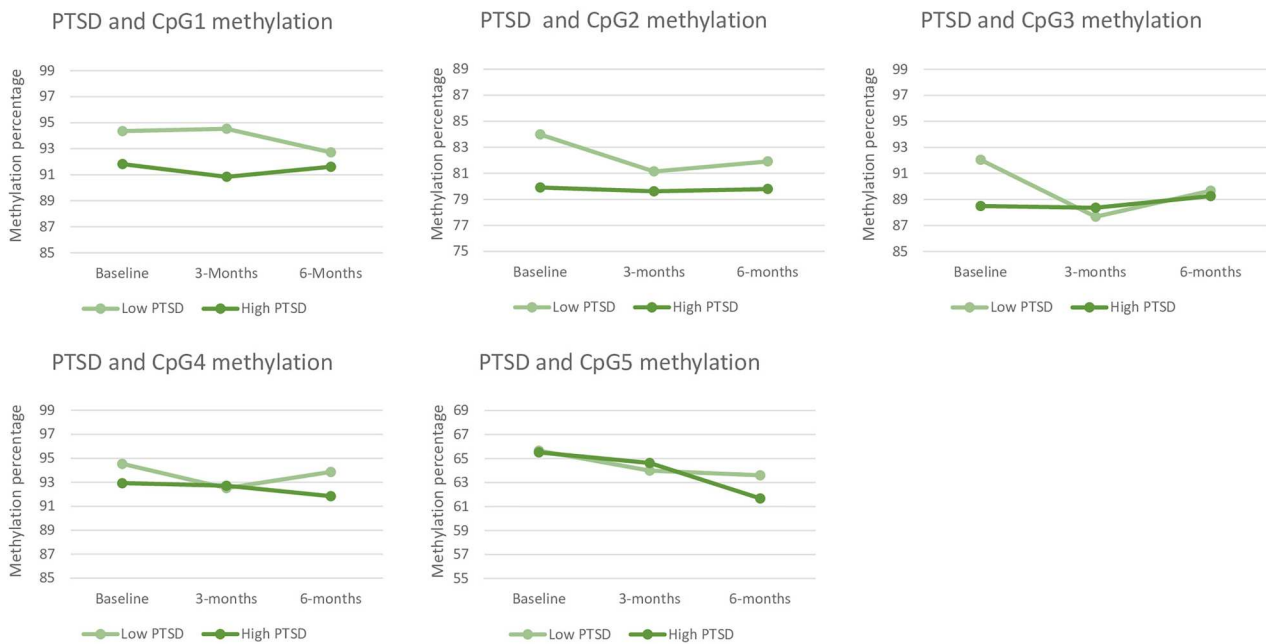


Figure 3. PTSD and differences in methylation.

Only one longitudinal study investigating PTSD and *FKBP5* methylation has previously been conducted. Similar to our findings, it was reported that higher *FKBP5* intron 7 methylation was associated with a faster PTSD recovery time in participants receiving a mindfulness-based intervention (Bishop et al., 2018).

A key finding of the study is the interaction effect between the rs1360780 genotypes and childhood trauma in predicting CpG1 methylation over time. In the CC genotype carriers, lower childhood trauma was associated with higher CpG1 methylation. This finding is in contrast with previous findings where the CT/TT genotype carriers with lower childhood trauma scores had higher *FKBP5* intron 7 methylation compared to CC carriers, although not all studies investigated the exact same sites (Grasso, 2020; Klenkel et al., 2013; Ramo-fernández et al., 2019; Yehuda et al., 2016b). Our findings suggest that the CC genotype carriers are more resilient when considering that lower PTSD scores were associated with higher CpG1 methylation.

Prior studies all followed a cross-sectional design. When considering our baseline findings there were no difference in methylation levels between CT/TT and CC genotypes. From the World Health Organisation's (WHO) multi-country mental health surveys, it is known that rape is the trauma that carries the highest conditional risk for PTSD (Koenen et al., 2017). The overwhelming, intrusive and often violent act of rape may result in a fight, flight, freeze or fawn response or a combination of responses with many women also experiencing dissociation, depersonalisation and derealisation during and after the event (Tschoeke et al., 2019). No previous studies have investigated methylation and rape solely, and

especially the effect of rape on methylation in the acute aftermath of the trauma. It is likely that the severely traumatic nature of rape may have such a profound effect on the HPA-axis that protective factors such as lower or no childhood trauma and rs1360780 genotype may be temporarily irrelevant (D'Elia et al., 2021).

Furthermore, it is likely that the effect of childhood trauma exposure may have resulted in a permanently altered HPA-axis in the CT/TT genotype which resulted in a stable association over time, between childhood trauma and methylation, in the CT/TT genotype carriers. The ability of the HPA-axis to change over time in the CC genotype may indicate a greater flexibility in recovery post-rape in this group (Niki Hosseini-Kamkar et al., 2021). However, the ability to draw comparisons between our findings and previous studies is limited by the absence of longitudinal studies with a similar design. Replication is needed to accurately interpret the relationship between childhood trauma, rs1360780 genotype and *FKBP5* intron 7 methylation.

In conclusion, the study provides evidence linking childhood trauma and the rs1360780 genotype to *FKBP5* methylation, but not directly to PTSD symptom scores, over a period of 6-months after experiencing a rape. It is plausible that a prolonged increase in circulating cortisol subsequent to childhood trauma, coupled with the rs1360780 genotype, may result in permanently altered neuroendocrinological changes underlying HPA-axis function (Wang et al., 2018). However, further investigation and replication of these findings is warranted to support this hypothesis.

The findings reported in this study should be interpreted in light of some limitations. First, our sample

Table 4. Associations between PTSD score, methylation levels and potential covariates at baseline.

	Age (baseline)		Basic education completed – no vs yes (baseline)		Employed – no vs yes (baseline)		In a relationship – no vs yes (baseline)		HIV status – negative vs positive (baseline)		Smoker – no vs yes (baseline)		Lifetime trauma (baseline)		Alcohol use (baseline)		Medication – no vs yes (baseline)	
	r	p	z	p	z	p	z	p	z	p	z	p	r	p	r	p	z	p
PTSD score (baseline)	-.199	.052	-1.087	.291	-2.676	.007	-2.282	.022	-1.07	.284	-0.49	.626	.062	.550	-.067	.514	-1.46	.144
CpG1 (baseline)	-.064	.566	-1.087	.277	-.504	.614	-.677	.498	-.038	.702	-0.53	.598	.071	.524	.145	.188	-1.28	.200
CpG2 (baseline)	-.200	.068	-1.175	.084	-1.174	.240	-.457	.647	-.002	.982	-1.06	.290	.007	.947	.065	.557	-1.18	.239
CpG3 (baseline)	-.103	.353	-1.012	.312	-.006	.996	-1.045	.296	-1.08	.282	-0.80	.423	.008	.941	.005	.966	-0.01	.989
CpG4 (baseline)	.020	.859	-.717	.473	-.568	.570	-.077	.938	-0.24	.811	-0.87	.384	-.087	.430	-.088	.423	-0.44	.663
CpG5 (baseline)	-.202	.063	-1.068	.285	-.608	.543	-1.770	.077	-0.77	.441	-1.35	.179	.028	.799	-.002	.984	-0.65	.514

Abbreviations: Posttraumatic stress disorder (PTSD).

size was small, thus the study was statistically underpowered. Second, although substantial evidence exists linking PTSD to *FKBP5* genotype and differential methylation, PTSD remains a complex disease and its aetiology and trajectory is likely due to an interaction of multiple genes of small effect and several biological pathways. Third, whole blood was used to investigate differential methylation and not brain tissue, which is more likely to be a proximal marker of PTSD pathophysiology. However, *FKBP5* intron 7 methylation levels in whole blood have been found to correlate with methylation levels in brain tissue in animal studies (Ewald et al., 2014). Unfortunately, the existing databases linking blood-brain methylation correlations are based on Illumina CpG identifiers and only one CpG site (CpG3) investigated in this study is represented on the Illumina EPIC array. However, blood is easily accessible and blood biomarkers of PTSD risk could be used to prioritise treatment for individuals most likely to develop PTSD following trauma exposure.

Despite these limitations, there are notable study strengths. The rs1360980 genotype and the CpG sites in and around GREs investigated in this study have been associated with functional effects of the gene (Matosin et al., 2018). Therefore, investigating these sites contributes to the understanding of molecular mechanisms following trauma exposure. We investigated a relatively homogenous sample, both in terms of demographic characteristics (young women from low socio-economic conditions, and from the same ethnicity group) and trauma exposure, enabling us to control various confounding factors. This is the second study to investigate longitudinal changes in methylation levels in intron 7 of the *FKBP5* gene, but the first to investigate longitudinal changes in African women who experienced rape. The inclusion of rs1360980 in the study is of importance since SNPs may have ethnicity-dependent functional effects (Galanter et al., 2017). Rape carries a high conditional risk for PTSD and investigating the underlying molecular mechanisms contributing to recovery post-rape could guide future therapeutic

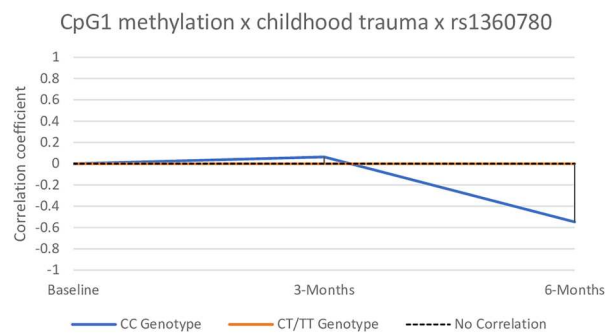
Table 5. Univariate associations of baseline confounding/covarying variables with rs1360980 CC vs CT/TT genotype.

	rs1360980 CC n = 31		rs1360980 CT/ TT n = 64		Comparison by genotype		
	M	SD	M	SD	n	z	P
Age	25.29	6.10	25.23	5.13	95	-0.25	.802
Lifetime trauma	1.39	1.56	1.38	1.28	95	-0.41	.680
Alcohol use	1.61	2.36	1.73	2.39	95	-0.22	.827
	n	%	n	%	n	χ^2	p
HIV positive	13	41.94	32	50.00	95	0.55	.460
Smoker	4	12.90	8	12.50	95	0.00	.956
Medication use	9	29.03	23	35.94	95	0.45	.504

Abbreviations: Posttraumatic stress disorder (PTSD).

Table 6. Mixed regression models testing the relationships between childhood trauma, rs1360780 genotype, *FKBP5* methylation and PTSD scores.

Model		β	Std error	<i>t</i>	<i>p</i>	95% CI Lower Upper	
1	<i>FKBP5</i> CpG1 methylation						
	Childhood trauma score	.145	.374	.387	.700	-.598	.887
	rs1360780 (CC vs CT/TT)	30.744	10.107	3.042	.003**	-.598	.887
	Childhood trauma score x rs1360780 (CC vs CT/TT)	-2.130	.588	-3.623	.000472**	-3.297	-.963
2	<i>FKBP5</i> CpG2 methylation						
	Childhood trauma score	-.316	.230	-1.378	.171	-.772	.139
	rs1360780 (CC vs CT/TT)	-12.481	6.288	-1.985	.050	-24.946	-.017
	Childhood trauma score x rs1360780 (CC vs CT/TT)	.648	.368	1.760	.081	-.082	1.379
3	<i>FKBP5</i> CpG3 methylation						
	Childhood trauma score	-.040	.244	-.163	.871	-.524	.444
	rs1360780 (CC vs CT/TT)	-16.120	6.660	-2.420	.017*	-29.316	-2.924
	Childhood trauma score x rs1360780 (CC vs CT/TT)	.855	.390	2.193	.030*	.083	1.627
4	<i>FKBP5</i> CpG4 methylation						
	Childhood trauma score	-.125	.161	-.774	.441	-.444	.195
	rs1360780 (CC vs CT/TT)	-9.040	4.487	-2.015	.046*	-17.927	-.153
	Childhood trauma score x rs1360780 (CC vs CT/TT)	.482	.264	1.824	.071	-.041	1.004
5	<i>FKBP5</i> CpG5 methylation						
	Childhood trauma score	-.122	.240	-.509	.612	-.600	.355
	rs1360780 (CC vs CT/TT)	6.452	6.756	.955	.342	-6.943	19.847
	Childhood trauma score x rs1360780 (CC vs CT/TT)	-.280	.396	-.706	.482	-1.066	.506
6	PTSD score over time						
	Childhood trauma score	.221	1.095	.203	.840	-1.955	2.399
	rs1360780 (CC vs CT/TT)	7.868	10.620	.741	.462	-13.377	29.112
	Childhood trauma score x rs1360780 (CC vs CT/TT)	-0.247	.663	-.373	.710	-1.567	1.072
7A	PTSD score over time						
	CpG1 methylation	-.335	.174	-1.921	.057	-.680	.010
7B	PTSD score over time						
	CpG1 methylation	-.324	.182	-1.784	.077	-.684	.036
	Childhood trauma score	.685	.764	.896	.372	-.831	2.201
	rs1360780 (CC vs CT/TT)	-12.951	21.39	-0.613	.542	-54.898	28.997
	Childhood trauma score x rs1360780 (CC vs CT/TT)	.646	1.257	.514	.608	-1.848	3.139
8A	PTSD score over time						
	CpG2 methylation	-.196	.224	-.875	.383	-.639	.246
8B	PTSD score over time						
	CPG2 methylation	-.230	.230	-1.002	.318	-.684	.224
	Childhood trauma score	.561	.762	.736	.463	-.950	2.073
	rs1360780 (CC vs CT/TT)	-16.759	21.316	-.786	.434	-59.031	25.514
	Childhood trauma score x rs1360780 (CC vs CT/TT)	1.011	1.254	.806	.422	-1.479	3.498
9A	PTSD score over time						
	CpG3 methylation	.018	.183	.101	.920	-.343	.379
9B	PTSD score over time						
	CpG3 methylation	-.037	.184	-.203	.839	-.401	.326
	Childhood trauma score	.614	.770	.798	.427	-.912	2.140
	rs1360780 (CC vs CT/TT)	-17.817	21.377	-.833	.407	-60.216	24.581
	Childhood trauma score x rs1360780 (CC vs CT/TT)	1.028	1.261	.815	.417	-1.473	3.529
10A	PTSD score over time						
	CpG4 methylation	-.055	.270	-.203	.840	-.588	.479
10B	PTSD score over time						
	CpG4 methylation	-.089	.271	-.327	.744	-.625	.447
	Childhood trauma score	.638	.778	.820	.414	-.904	2.181
	rs1360780 (CC vs CT/TT)	-21.979	21.644	-1.015	.312	-64.892	20.934
	Childhood trauma score x rs1360780 (CC vs CT/TT)	1.325	1.2815	1.034	.303	-1.215	3.865
11A	PTSD score over time						
	CpG5 methylation	.180	.207	.866	.388	-.230	.589
11B	PTSD score over time						
	CpG5 methylation	.130	.214	.608	.544	-.292	.553
	Childhood trauma score	.646	.766	.843	.401	-.874	2.165
	rs1360780 (CC vs CT/TT)	-17.305	21.100	-.820	.414	-59.151	24.542
	Childhood trauma score x rs1360780 (CC vs CT/TT)	.9848	1.247	.789	.432	-1.489	3.458

**Figure 4.** Correlation coefficients between childhood trauma and *FKBP5* CpG1 methylation, stratified by rs1360780 genotype, plotted over time.

interventions and thereby reduce the disease burden associated with rape.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available on request from the authors.

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